Chapter 11 Energy in Thermal Processes

Molar Specific Heat at Constant Volume

$$Q = nC_V\Delta T \text{ (constant volume)}$$

where C_V is a constant called molar specific heat at constant volume, Q is the heat for gas to increase ΔT temperature.

By the first law,

$$\Delta E_{\text{int}} = Q - W = nC_V\Delta T - 0 = nC_V\Delta T$$

Thus

$$C_V = \frac{\Delta E_{\text{int}}}{n\Delta T} = \frac{3}{2}R$$

Molar Specific Heat at Constant Pressure

$$Q = nC_p\Delta T \text{ (constant pressure)}$$

where C_p is a constant called molar specific heat at constant pressure.

By the first law,

$$\Delta E_{\text{int}} = Q - W = Q - p\Delta V = Q - nR\Delta T = n\Delta T(C_p - R)$$

Thus

$$C_V = C_p - R$$

The former induction is for monatomic ideal gas.

Adiabatic Expansion of Ideal Gas

Conditions:

- Happen very fast
- In a well insulated container(no heat transfer)

There exists

$$pV^\gamma = \text{a constant (adiabatic process)}$$

where $\gamma = C_p/C_V$.
That is,

$$p_i V_i^\gamma = p_f V_f^\gamma$$

since $pV = nRT$, we have

$$T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1} \text{ (abiabatic process)}$$

Free Expansion

Conditions:
The work is not done and the internal energy does not change.

Thus,

$$p_i V_i = p_f V_f \text{ (free expansion)}$$

Four Special Processes

$$\Delta E_{\text{int}} = Q - W = nC_V \Delta T$$

is suitable for every process.

Process Type	Constant Quantity	Special Results
Isobaric	P	$Q = nC_p \Delta T; W = p\Delta V$
Isothermal	T, pV	$Q = W = nRT \ln\left(\frac{V_f}{V_i}\right); \Delta E_{\text{int}} = 0$
Adiabatic	$pV^\gamma, TV^{\gamma-1}$	$Q = 0; W = -\Delta E_{\text{int}}$
Isochoric	V	$W = 0; Q = \Delta E_{\text{int}} = nC_V \Delta T$
free expansion	T, pV	$\Delta E_{\text{int}} = Q = W = 0$

Change in Entropy

$$\Delta S = S_f - S_i = \int_i^f \frac{1}{T}\, dQ \text{ (change in entropy defined)}$$

For isothermal expansion

$$\Delta S = \frac{1}{T} \int_i^f dQ = \frac{Q}{T}$$

By the first law,

$$
\begin{aligned}
dE_{\text{int}} &= dQ - dW \\
dQ &= pdV + nC_V dT \\
\frac{dQ}{T} &= nR\frac{dV}{V} + nC_V\frac{dT}{T} \\
\int_i^f \frac{1}{T}\,dQ &= \int_i^f nR\frac{1}{V}\,dV + \int_i^f nC_V\,\frac{dT}{T}
\end{aligned}
$$

Thus,

$$\Delta S = S_f - S_f = nR\ln\frac{V_f}{Vi} + nC_V\ln\frac{T_f}{T_i}$$

Entropy and Disorder

Entropy can be described in terms of disorder. And a disorderly arrangement is more probable than orderly one without interference.

$$S = k\ln W$$

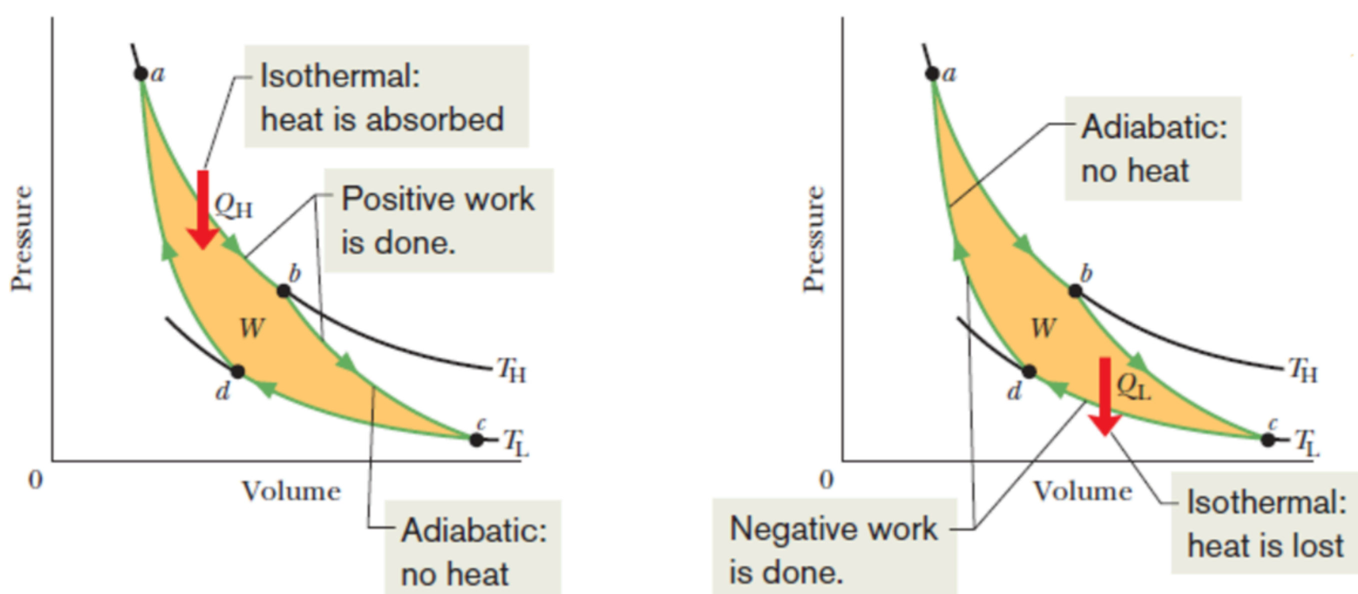
where W is multiplicity of configuration. k is Boltzmann constant.

Carnot Engine

A heat engine operating in an ideal, reversible cycle between two reservoirs.

Carnot's Theorem:
No real engine operating between two energy reservoirs can be more efficient than a Carnot engine operating between the same two energy reservoirs.
Or, Carnot engine is the most efficient engine.

Carnot Cycle



(a) (b)

a->b: Isothermal, heat is absorbed and work is positive
b->c: Adiabatic, no heat, temperature decreases and work is positive
c->d: Isothermal, heat is lost and work is negative
d->a: Adiabatic, no heat, temperature increases and work is negative

Entropy Changes

$$\Delta S = 0(\text{reversible process}) = \Delta S_H + \Delta S_L = \frac{|Q_H|}{T_H} - \frac{|Q_L|}{T_L}$$

thus

$$\frac{|Q_H|}{T_H} = \frac{|Q_L|}{T_L}$$

Efficiency of a Carnot Engine

$$\epsilon = \frac{\text{energy we get}}{\text{energy we pay for}} = \frac{|W|}{|Q_H|}$$

Since

$$W = |Q_H| - |Q_L|$$

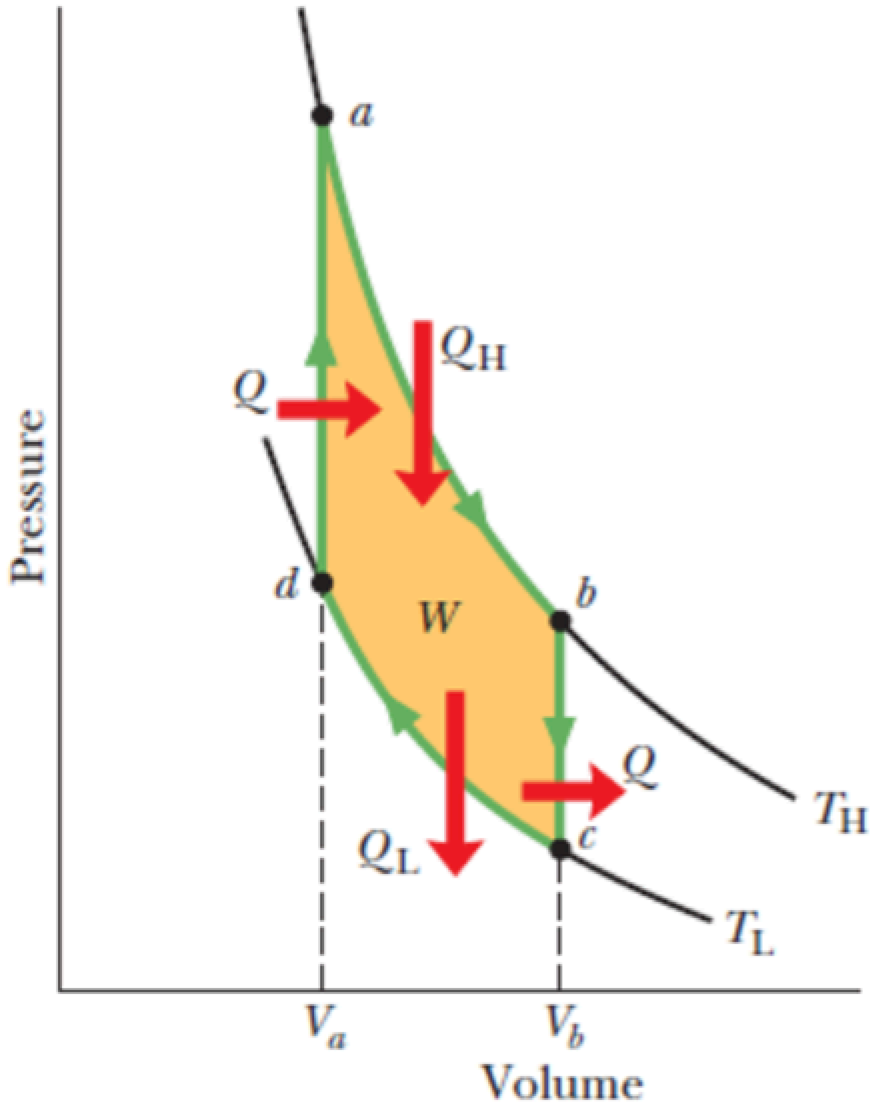
and

$$\frac{|Q_H|}{T_H} = \frac{|Q_L|}{T_L}$$

Thus

$$\epsilon_c = 1 - \frac{T_L}{T_H}$$

Stirling Engine



a->b: isothermal
b->c: isochoric
c->d: isothermal
d->a: isochoric

The efficiency of ideal Stirling engine is lower than Carnot engine.

Refrigerators

Use work to transfer energy from a low temperature reservoir to a high-temperature reservoir. The efficiency is

$$K = \frac{\text{what we want}}{\text{what we pay for}} = \frac{|Q_L|}{|W|}$$

Similiarly,

$$K_C = \frac{T_L}{T_H - T_L}$$